Xixian Niu

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EDUCATION

Ph.D. in Economics, Université Libre de Bruxelles (ECARES and I3h) expected 2027
Supervisors: Micael Castanheira and Mathias Dewatripont

Research Master in Economics, ECARES, Université Libre de Bruxelles 2023
Supervisor: Mathias Dewatripont

M.Sc. in Quantitative Economics (QEM), Paris 1 Panthéon-Sorbonne and UCLouvain 2022
Supervisor: Hervé Moulin

B.Sc. in Mathematical Statistics, School of Economics, Xiamen University 2018

RESEARCH FIELDS

Primary: Economics of Innovation, Health Economics, Industrial Organization
Secondary: Applied Microeconomics, Applied Econometrics

JOB MARKET PAPER

"Asymmetric Intellectual Property Rights and Follow-On Innovation: Evidence from Drug Repurposing"

Patent protection over a redeployed technology is vertically asymmetric: a follow-on innovator's patent reaches only the new use, never the technology beneath it. The broad upstream patent is therefore not merely a gate the follow-on licenses through, but a shield it sells behind, since imitating the follow-on infringes the upstream patent for as long as it stands. Longer upstream protection can consequently raise follow-on innovation rather than deter it, reversing the prediction of the symmetric benchmark in the literature. Using a new dataset of all FDA-approved new molecular entities, 1985–2024, and instrumenting exclusivity with the patent-filing-to-trial gap, I estimate that an additional year of protection raises expected repurposing by roughly twenty percent, with the effect strongest precisely where the shield rather than the gate should bind.

PUBLICATIONS

S. Akodad, **X. Niu**, B. Secades, and H. Stevens. "Impact of Drug Repurposing between 1985 and 2024 on Pharmaceutical Innovation." *Communications Medicine*, 6:84, 2026. doi.org/10.1038/s43856-025-01344-1

WORKING PAPERS

"Do Policy Incentives Align with Pharmaceutical Innovation Challenges? A Retrospective Analysis of 42 Countries, 2000–2022." With S. Mermelstein, A. Gentilini, C. Yue, H. Stevens, M. Shaikh, and M. Miraldo. Invited for a special issue of *Health Economics*.

We assemble the first systematic dataset of 1,078 pharmaceutical innovation policies across 42 countries and 163 diseases (2000–2022), decomposing each policy into underlying features to move beyond the conventional push–pull distinction. Push and pull incentives prove similarly allocated, both concentrating in scientifically difficult areas with heavy academic and small-firm involvement. At the feature level, however, allocation diverges: financial and risk-sharing incentives follow scientific difficulty, expedited regulatory pathways follow disease severity and treatment scarcity rather than total burden, and knowledge-pooling and access conditions remain narrowly used. Policy activity clusters in high-income settings and in cancers and rare diseases, leaving antimicrobial resistance and maternal and neonatal conditions persistently underserved.

"Unbundling Global Innovation Policies: Mechanism Effects, Spillovers, and the Equitable Allocation of Pharmaceutical R&D." With C. Yue, S. Mermelstein, M. Miraldo, H. Stevens, M. Shaikh, and M. Dewatripont.

Innovation policies bundle several incentive mechanisms with offsetting effects, so estimated policy-level effects identify no single margin. Decomposing 1,100 policy introductions into six mechanisms and linking them to 1.9 million R&D grants across 42 countries (2000–2021), we exploit within-policy variation in mechanism composition to separate own-country effects, cross-country spillovers, and allocation relative to disease burden. Revenue subsidies raise R&D but redirect it toward commercially lucrative diseases and high-income markets, widening the gap between private and social returns. Mechanisms that lower regulatory and information frictions yield larger and more persistent gains, concentrated where profit incentives are weakest, and generate positive spillovers across borders. Mandated IP-sharing lowers appropriability and depresses short-run entry.

Success-contingent rewards dominate unconditional transfers, consistent with reduced moral hazard and better selection.

WORK IN PROGRESS

"Patent Box Tax Relief and Orphan Drug Development."

Nine EU countries have extended preferential IP-box tax rates to orphan-drug income since 2015. Using difference-in-differences on a firm–country–year panel of 78,060 observations across 30 EMA member states (2000–2024), I find that adoption raises orphan-drug launches by roughly 75 percent. The effect is confined to large firms, with no response from SMEs and no increase in designations or registrations, indicating that the policy stimulates commercialization rather than innovation.

RESEARCH PROJECTS

Pharmaceutical innovation policy in 42 countries, Core Contributor Brussels · 2023 – 2026

Hi-Prix Work Package 7, co-led by ULB and Imperial College London (Horizon Europe, €5.3M, 18 partner organizations in 10 countries); doctoral researcher funded by the project

Output: "Incentives for Pharmaceutical Innovation and Equitable Access to Innovation." Policy report for Hi-Prix, 2026. With M. Miraldo, H. Stevens, C. Yue, S. Mermelstein, A. Gentilini, M. Shaikh, and M. Dewatripont.

RESEARCH EXPERIENCE

Visiting Researcher, Imperial College London May 2023 – May 2024

Centre for Health Economics and Policy Innovation, host: Prof. Marisa Miraldo. Empirical research on pharmaceutical innovation policy within Hi-Prix; co-authored the two working papers and the policy report listed above.

Research Intern, Fair Division project, Paris 1 Panthéon-Sorbonne 2021 – 2022

Supervisor: Prof. Hervé Moulin

PRESENTATIONS

2025 Hi-Prix Consortium Annual Conference, Brussels (poster)
2024 Hi-Prix Consortium Annual Conference, Lisbon

FELLOWSHIPS AND AWARDS

Ph.D. Fellowship, Horizon Europe Hi-Prix project 2023 – 2026

SKILLS AND LANGUAGES

Programming: Python, R, Stata, Quarto / R Markdown, LaTeX (proficient); SQL, Git (basics).
Data engineering: API-based collection (Dimensions, Cortellis); web scraping (Requests, BeautifulSoup, Selenium); PDF text extraction; LLM-assisted extraction from unstructured documents; record linkage and server-based workflows for datasets with millions of records.
Methods: Panel data, applied econometrics and causal inference, including difference-in-differences, instrumental variables (including shift-share), regression discontinuity, and synthetic control; industrial-organization modeling; machine learning for feature selection (LASSO, random forest); questionnaire design and expert interviews.
Domain data: FDA Orange Book and Patent Term Extension records; EMA orphan-drug designations; Cortellis Competitive Intelligence; Dimensions (publications, patents, grants, clinical trials); ICD-11; cross-country R&D tax, IP-box and HTA policy.
Languages: Chinese (native), English (fluent).

REFERENCES

Micael Castanheira, FNRS Research Director and Professor, ECARES, Université Libre de Bruxelles
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Mathias Dewatripont, Professor of Economics, ECARES, Université Libre de Bruxelles
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